Nonequilibrium Josephson Effect in SIS Junctions

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This document summarizes the equations solved numerically in the code and specifies the conventions used throughout the calculations.

I. MODEL

The nonequilibrium Josephson current in an SIS junction can be represented in a compact matrix form (for details see Ref. 1)

$$I(t) = \frac{e}{8}\,\mathrm{Sp}\Big\{[\mathbb{S}_n + \Gamma]^{-1}(H - \Gamma H \Gamma^+) \times [\mathbb{S}_n^+ + \Gamma^+]^{-1}(\mathbb{S}_n^+ \hat{\tau}_3 \hat{o}_3 \mathbb{S}_n - \hat{\tau}_3\hat{o}_3)\Big\}, \tag{1}$$

where I denotes the Josephson current carried by a single conducting channel that hosts quasiparticle states with both spin orientations, $\hat{\tau}_3$ and $\hat{o}_3$ are Pauli matrices in Nambu and trajectory spaces, respectively, and $\mathbb{S}_n$ is the normal-state interface scattering matrix.

For simplicity we assume that the interface preserves the quasiparticle spin projection. Then the scattering matrix for a given spin projection $\sigma = \pm 1$ has the form

$$[\mathbb{S}_n]_\sigma = \begin{pmatrix} \sqrt{R_\sigma} & -i\sqrt{D_\sigma} & 0 & 0 \\ -i\sqrt{D_\sigma} & \sqrt{R_\sigma} & 0 & 0 \\ 0 & 0 & \sqrt{R_{-\sigma}} & i\sqrt{D_{-\sigma}} \\ 0 & 0 & i\sqrt{D_{-\sigma}} & \sqrt{R_{-\sigma}} \end{pmatrix} e^{-i\sigma\theta/2}, \tag{2}$$

is the normal-state interface transmission probability for quasiparticles with spin projection σ, and θ is the spin-mixing angle.

For a given time-dependent voltage bias $V(t)$, the superconducting phase difference across the junction reads

$$\varphi(t) = 2e\int^t V(t')dt'. \tag{3}$$

The kernels of the operators Γ and H are then given by

$$\Gamma(t,t') = e^{i\varphi(t)\hat{\tau}_3\hat{o}_3/4}\Gamma_{eq}(t-t')e^{-i\varphi(t')\hat{\tau}_3\hat{o}_3/4}, \tag{4}$$

$$H(t,t') = e^{i\varphi(t)\hat{\tau}_3\hat{o}_3/4}H_{eq}(t-t')e^{-i\varphi(t')\hat{\tau}_3\hat{o}_3/4}, \tag{5}$$

where

$$H_{eq}(t-t') = \int_{-\infty}^{\infty}\frac{d\varepsilon}{2\pi}h_0(\varepsilon)e^{-i\varepsilon(t-t')}, \quad h_0(\varepsilon) = \tanh\frac{\varepsilon}{2\pi}, \tag{6}$$

$$\Gamma_{eq}(t-t') = \hat{\tau}_1\int_{-\infty}^{\infty}\frac{d\varepsilon}{2\pi}a(\varepsilon)e^{-i\varepsilon(t-t')}, \quad a(\varepsilon) = \frac{-\varepsilon + \sqrt{\varepsilon^2 - |\Delta|^2}}{|\Delta|}. \tag{7}$$

We assume that the superconducting order parameter is spatially uniform in both leads and has the same magnitude $|\Delta|$, corresponding to a symmetric junction.

For a time-independent voltage bias, evaluation of the current can be reduced to solving matrix recurrence relations derived in Ref. 1. These recurrence relations are implemented numerically in the present code.

For reference, in the limit of a transparent junction biased by a small voltage $|eV| \ll |\Delta|$ one obtain[2]

$$I = e|\Delta||\sin(eVt)|\tanh\frac{|\Delta|}{2T}\,\mathrm{sgn}\,V. \tag{8}$$

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[1] M.S. Kalenkov and A.D. Zaikin, Absence of fractional ac Josephson effect in superconducting junctions, Phys. Rev. B 110, 134521 (2024).

[2] D. Averin and A. Bardas, ac Josephson Effect in a Single Quantum Channel, Phys. Rev. Lett. **75**, 1831 (1995).